

234054F

Question 03
Part A
a)

$$Q_d = 100 - 2P_c$$

$$Q_s = 4P_p - 20$$

$$\text{Tax} = 15$$

$$P_c = P_p + 15$$

$$\begin{aligned} Q_d &= 100 - 2(P_p + 15) \\ &= 100 - 2P_p - 30 \\ &= 70 - 2P_p \end{aligned}$$

$$Q_d = Q_s$$

$$70 - 2P_p = 4P_p - 20$$

$$90 = 6P_p$$

$$P_p = 15$$

$$P_c = P_p + 15$$

$$= 15 + 15$$

$$P_c = 30$$

$$Q = 4(15) - 20$$

$$Q = 40$$

$$\text{Quantity} = 40$$

$$\text{Price paid by Consumers} = 30$$

$$\text{Price received by Producers} = 15$$

b)

$$\begin{aligned}\text{Tax revenue} &= \text{Tax} \times \text{quantity} \\ &= 15 \times 40 = 600\end{aligned}$$

Before tax, Price = 20

after tax,

Consumers pay 30 $\rightarrow$ +10

Producers receive 15 $\rightarrow$ -5

$\therefore$ Consumers bear larger burden (10)

Producers bear smaller burden (5)

Part B

a) Before tax

Demand intercept

$$Q_d = 100 - 2P \rightarrow P = 50$$

Supply intercept

$$Q_s = 4P - 20 \rightarrow P = 5$$

Producer Surplus (PS)

$$PS = \frac{1}{2} (20 - 5) (60)$$

$$= \frac{1}{2} (15 \times 60)$$

$$= 450$$

Total welfare (before)

$$TW = 900 + 450$$

$$= 1350$$

Consumer surplus (CS)

$$\begin{aligned}CS &= \frac{1}{2} (50 - 20) (60) \\ &= 900\end{aligned}$$

after tax

$$CS = \frac{1}{2} (50 - 30) (40)$$

$$= \frac{1}{2} (20 \times 40)$$

$$= 400$$

$$PS = \frac{1}{2} (15 - 5) (40)$$

$$= 200$$

$$\text{Tax revenue} = 600$$

Total welfare (after)

$$TW = 400 + 200 + 600$$

$$= 1200$$

b) Deadweight loss = $1350 - 1200$
= 150

Tax reduces quantity from 60 $\rightarrow$ 40

Some mutually beneficial trades no longer occurs.

This leads to loss of economic efficiency.

Part C

a) Demand is relatively inelastic

Consumers cannot easily reduce consumption.

Consumers bear more tax burden

Producers bear less

Because less elastic side bears more burden.

b) Economic efficiency

Tax creates deadweight loss (150)

$\therefore$ Not fully efficient.

Social appropriateness.

* This is life saving antibiotic

* Demand is inelastic.

Essential good.